

Appendix

Learning by Shifting: Temporal View Construction for Time Series Contrastive Learning

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Code Availability – To support reproducibility, the complete codebase and scripts for ShiFT are provided in the main repository <https://github.com/sfi-norwai/ShiFT>.

A Reproducibility and Computational Environment

To ensure reproducibility, we report all experimental settings and computational details used in our study. Experiments on the large-scale benchmarks are conducted over five random seeds $\{1, 2, 3, 4, 5\}$, and results are reported as averages across seeds. For the UCR and UEA benchmarks, we follow prior work and use a fixed random seed of 1. ShiFT uses a contrastive temperature of $\tau = 0.1$ and overlap $\rho = 0.5$ throughout.

All experiments are implemented in Python 3.12.3 using PyTorch 2.5.1 with CUDA 12.4. Experiments are executed on NVIDIA Tesla V100S GPUs with 32 GB memory. The system uses NVIDIA driver version 570.172.08.

B Baseline Descriptions.

This appendix provides implementation details and reproduction settings for all baseline methods compared in this work. To ensure a controlled and fair comparison, we standardize the backbone architecture and data-loading pipeline across all methods. Specifically, all models are trained using a shared 1D InceptionTime backbone [5], ensuring that performance differences reflect contrastive learning objectives rather than architectural choices. All models are optimized using AdamW with weight decay of 1×10^{-4} and momentum parameters $\beta_1 = 0.9$, $\beta_2 = 0.99$, under a cosine learning-rate schedule with linear warm-up during the first 10% of training iterations.

Learning rates are selected via a hyperparameter search over $\{1 \times 10^{-3}, 3 \times 10^{-4}, 1 \times 10^{-4}\}$ on the six large-scale datasets. Contrastive methods (InfoTS, TS2Vec, SimCLR, and ShiFT) perform best at 1×10^{-3} , while the masked modelling method SimMTM performs best at 1×10^{-4} . Coincidentally, these

values correspond to the default learning rates reported in the respective original implementations. For the UCR and UEA benchmarks, where a batch size of 8 is used instead of the default 128, learning rates are adjusted proportionally following the linear scaling rule:

$$\text{lr} = \text{lr}_{\text{base}} \times \frac{8}{128}, \quad (1)$$

yielding 6.25×10^{-5} for contrastive methods and 6.25×10^{-6} for SimMTM. Unless otherwise stated, all other method-specific hyperparameters are taken directly from the official implementations released by the original authors, with no dataset-specific tuning performed.

InfoTS [7]. We use the official implementation with default method-specific hyperparameters as described in the original work, including the combined global and local contrastive objectives.

TS2Vec [10]. We adopt the official implementation with default settings.

SimCLR [2]. SimCLR was originally proposed for computer vision; we use a time series adaptation sourced from a public repository in [6], applying jittering ($\sigma = 0.5$) and scaling ($\sigma = 0.3$) as data augmentations, consistent with prior works in time series contrastive learning [7,9].

SimMTM [4]. We use the official implementation with default masking and training configurations.

C Detailed Dataset Information.

We use publicly available time series datasets previously employed in self-supervised representation learning research. The WISDM2, PAMAP2, SLEEP and SKODA datasets were directly obtained from the repository linked by the Series2Vec project, without applying additional preprocessing. For the ECG dataset, we generate random windows for self-supervised learning following [8]. Each subject’s continuous 12-hour recording was split into 10-minute segments to reduce redundancy, and during training, a random window of 2,500 time steps was sampled from each segment. The first 5 subjects were held out for testing, with the remaining subjects used for training. This strategy increases variability in the training data while maintaining consistent input dimensions across subjects. For linear probing and kNN evaluation, we extract a deterministic window of 2,500 time steps from the middle of each segment from the train set to ensure reproducible evaluation. For the HARTH dataset, we performed subject-wise split with 22 subjects in total. The first 18 subjects were used for training and the remaining 4 for testing. Each subject’s multivariate time series was split into overlapping windows of 500 time steps with a stride of 250, and the majority class within each window was assigned as the label.

Table 1 summarizes the key characteristics of each dataset. All datasets are split using subject-wise splitting, the SKODA dataset, collected from a single subject, however, does not have subject-wise splits.

Table 1: Summary of datasets used in the large-scale experiments.

Dataset	Train set	Test set	Subject	Sequence Length	Channel	Class
ECG	59,922	16,645	23	2,500	2	4
HARTH	23,025	2,785	22	500	6	18
PAMAP2	51,192	11,590	9	100	52	18
SKODA	22,587	5,646	1	50	64	11
SLEEP	25,612	8,910	20	3,000	1	5
WISDM2	134,614	14,421	29	40	3	6

D Full UCR & UEA Benchmarks Results.

Table 3 and Table 2 present the full UCR and UEA benchmark results respectively. The UEA archive [1] originally contains 30 datasets. In our experiments, we report results on 28 datasets, excluding *InsectWingbeat* and *EigenWorms*. These two datasets exhibit atypical input shapes and extremely large memory requirements, which lead to runtime or memory errors for several baseline methods despite consistent preprocessing and hyperparameter settings. The UCR archive [3] contains 128 datasets. We evaluate on 124 datasets, excluding *DodgerLoopDay*, *DodgerLoopGame*, *DodgerLoopWeekend*, and *MelbournePedestrian*, which consistently produce NaN values or runtime errors across multiple baselines despite standard normalization and training procedures.

We note that this behavior is not unique to our evaluation protocol. Prior work such as TS2Vec [10] similarly omits a comparable subset of datasets, reporting results on 125 UCR and 29 UEA datasets, due to analogous issues related to input dimensionality, memory constraints, or training instability.

Table 2: 28 UEA - Performance across datasets. Best per dataset in bold, second best underlined.

dataset	InfoTS	Rand Init.	ShiFT	SimCLR	SimMTM	TS2Vec
ArticularyWordRecognition	95.67	96.33	98.33	<u>97.33</u>	96.67	97.00
AtrialFibrillation	<u>26.67</u>	<u>26.67</u>	33.33	<u>26.67</u>	<u>26.67</u>	<u>26.67</u>
BasicMotions	100.00	100.00	100.00	100.00	100.00	100.00
CharacterTrajectories	97.98	97.98	99.44	<u>99.37</u>	99.23	99.30
Cricket	98.61	98.61	98.61	97.22	95.83	97.22
DuckDuckGeese	58.00	60.00	60.00	54.00	50.00	54.00
ERing	72.59	80.37	88.89	<u>89.63</u>	88.89	91.48
Epilepsy	92.03	93.48	<u>94.93</u>	96.38	94.20	93.48
EthanolConcentration	31.18	<u>30.80</u>	22.05	26.24	23.95	28.52
FaceDetection	52.27	52.67	<u>53.52</u>	55.02	52.01	52.38
FingerMovements	46.00	56.00	57.00	<u>59.00</u>	62.00	56.00
HandMovementDirection	43.24	<u>35.14</u>	29.73	25.68	29.73	33.78
Handwriting	36.00	38.71	47.18	51.06	46.47	<u>50.59</u>
Heartbeat	<u>73.17</u>	74.15	71.71	72.68	71.71	72.68
JapaneseVowels	97.57	96.49	97.57	97.30	97.57	96.76
LSST	<u>51.42</u>	50.49	51.34	51.30	51.62	50.12
Libras	58.33	60.56	88.33	76.67	82.22	<u>85.00</u>
MotorImagery	54.00	54.00	<u>55.00</u>	48.00	51.00	61.00
NATOPS	82.78	83.89	92.78	<u>89.44</u>	88.33	88.33
PEMS-SF	78.61	<u>77.46</u>	71.10	72.83	73.41	73.41
PenDigits	96.86	97.03	96.91	97.63	97.88	<u>97.77</u>
PhonemeSpectra	19.68	19.33	25.68	<u>25.26</u>	24.66	24.55
RacketSports	81.58	83.55	81.58	81.58	<u>82.89</u>	81.58
SelfRegulationSCP1	84.30	<u>86.01</u>	84.98	85.32	87.03	84.30
SelfRegulationSCP2	53.89	48.89	46.67	<u>50.56</u>	48.33	46.67
SpokenArabicDigits	93.72	93.36	97.14	97.14	97.54	<u>97.45</u>
StandWalkJump	13.33	33.33	<u>40.00</u>	26.67	46.67	33.33
UWaveGestureLibrary	58.44	58.44	83.44	<u>81.88</u>	81.56	81.25
Average	66.00	67.28	70.26	68.99	69.58	69.81

Table 3: 124 UCR Performance across datasets. Best per dataset in bold, second best underlined.

dataset	InfoTS	Rand Init.	ShiFT	SimCLR	SimMTM	TS2Vec
ACSF1	83.00	78.00	93.00	<u>86.00</u>	<u>86.00</u>	<u>86.00</u>
Adiac	71.10	72.63	83.12	79.54	<u>80.05</u>	77.24
AllGestureWiimoteX	46.14	49.86	<u>70.14</u>	68.43	70.57	70.00
AllGestureWiimoteY	52.14	52.71	<u>73.00</u>	71.14	71.14	74.14
AllGestureWiimoteZ	50.57	47.00	74.43	<u>71.43</u>	68.43	67.43
ArrowHead	72.57	75.43	83.43	79.43	80.00	<u>80.57</u>
BME	82.67	65.33	98.00	94.00	92.00	98.00
Beef	60.00	60.00	<u>66.67</u>	60.00	63.33	73.33
BeetleFly	75.00	80.00	95.00	80.00	80.00	<u>90.00</u>
BirdChicken	75.00	80.00	90.00	100.00	90.00	100.00
CBF	96.67	89.11	<u>99.00</u>	98.89	99.56	<u>99.00</u>
Car	61.67	55.00	83.33	<u>80.00</u>	<u>80.00</u>	73.33
Chinatown	94.75	95.04	96.21	97.67	95.34	<u>97.38</u>
ChlorineConcentration	58.59	57.50	64.79	<u>66.02</u>	65.89	67.27
CinCECGTorso	53.48	47.46	72.68	58.41	58.19	<u>58.77</u>
Coffee	78.57	96.43	100.00	100.00	100.00	96.43
Computers	68.40	70.80	81.20	75.60	76.80	<u>77.20</u>
CricketX	64.62	64.62	<u>73.33</u>	72.31	74.87	73.08
CricketY	57.95	60.00	73.08	<u>72.05</u>	71.79	69.23
CricketZ	65.13	65.64	78.72	76.41	<u>77.18</u>	75.38
Crop	60.63	60.89	72.39	<u>72.96</u>	73.01	72.46
DiatomSizeReduction	84.31	84.64	<u>86.93</u>	88.24	86.27	<u>86.93</u>
DistalPhalanxOutlineAgeGroup	<u>73.38</u>	70.50	72.66	72.66	72.66	75.54
DistalPhalanxOutlineCorrect	76.45	77.54	77.90	76.81	77.54	77.90
DistalPhalanxTW	68.35	65.47	<u>67.63</u>	66.19	<u>67.63</u>	66.91
ECG200	85.00	81.00	87.00	89.00	<u>90.00</u>	91.00
ECG5000	93.42	93.56	93.56	94.00	<u>93.78</u>	93.62
ECGFiveDays	85.95	87.46	100.00	98.14	87.80	<u>98.95</u>
EOGHorizontalSignal	46.41	51.93	57.46	50.83	<u>54.70</u>	51.38
EOGVerticalSignal	39.50	42.82	39.78	37.29	<u>41.16</u>	37.29
Earthquakes	<u>76.98</u>	74.82	75.54	74.10	74.10	78.42
ElectricDevices	67.76	<u>67.60</u>	66.42	66.41	66.62	66.79
EthanolLevel	52.60	50.60	52.80	54.60	<u>60.00</u>	61.00
FaceAll	83.85	89.64	<u>90.77</u>	88.40	91.07	90.12
FaceFour	80.68	82.95	<u>94.32</u>	87.50	89.77	95.45
FacesUCR	84.44	80.88	92.54	90.24	91.17	92.54
FiftyWords	50.33	49.01	73.41	75.60	74.29	<u>75.16</u>
Fish	78.29	81.14	<u>96.57</u>	94.29	97.14	94.86
FordA	93.48	92.95	<u>93.86</u>	94.24	93.03	93.18
FordB	81.73	<u>80.86</u>	77.78	79.01	80.00	79.01
FreezerRegularTrain	93.58	91.58	98.67	<u>98.32</u>	98.25	97.65
FreezerSmallTrain	77.54	82.46	75.79	84.42	75.47	<u>83.09</u>
Fungi	73.12	73.12	97.31	90.86	<u>91.40</u>	<u>91.40</u>
GestureMidAirD1	60.77	60.77	<u>65.38</u>	60.77	66.15	64.62
GestureMidAirD2	53.08	40.77	<u>54.62</u>	51.54	55.38	51.54
GestureMidAirD3	27.69	<u>30.00</u>	31.54	22.31	27.69	26.15
GesturePebbleZ1	80.81	72.09	87.79	83.72	83.14	<u>85.47</u>
GesturePebbleZ2	79.75	74.05	85.44	81.01	<u>83.54</u>	81.65
GunPoint	96.00	96.00	99.33	<u>98.67</u>	96.00	98.00
GunPointAgeSpan	60.13	52.22	99.68	96.20	98.10	<u>98.42</u>
GunPointMaleVersusFemale	58.86	62.97	100.00	96.52	<u>99.68</u>	99.05
GunPointOldVersusYoung	100.00	100.00	100.00	100.00	99.37	100.00
Ham	65.71	71.43	<u>75.24</u>	76.19	<u>75.24</u>	71.43
HandOutlines	78.92	78.38	92.16	87.30	92.16	89.19
Haptics	38.64	40.26	<u>47.73</u>	49.35	47.40	46.43
Herring	59.38	<u>65.62</u>	67.19	64.06	62.50	62.50
HouseTwenty	94.96	92.44	<u>93.28</u>	87.39	83.19	82.35

InlineSkate	25.27	25.64	37.45	<u>27.09</u>	26.18	<u>27.09</u>
InsectEPGRegularTrain	100.00	100.00	100.00	100.00	100.00	100.00
InsectEPGSmallTrain	100.00	100.00	100.00	100.00	100.00	100.00
InsectWingbeatSound	43.89	41.21	60.45	56.21	56.92	<u>57.22</u>
ItalyPowerDemand	95.92	95.14	95.53	95.14	96.70	<u>96.21</u>
LargeKitchenAppliances	84.27	86.13	83.73	85.60	86.13	85.07
Lightning2	72.13	78.69	78.69	75.41	72.13	77.05
Lightning7	60.27	64.38	<u>78.08</u>	<u>78.08</u>	84.93	<u>78.08</u>
Mallat	84.48	82.52	<u>92.45</u>	91.47	95.91	91.56
Meat	93.33	93.33	86.67	91.67	93.33	91.67
MedicalImages	69.08	66.32	76.32	<u>78.82</u>	76.58	81.18
MiddlePhalanxOutlineAgeGroup	59.09	<u>55.84</u>	48.05	55.19	50.00	51.95
MiddlePhalanxOutlineCorrect	79.38	81.10	79.04	79.38	<u>80.76</u>	80.41
MiddlePhalanxTW	50.65	51.30	53.25	51.95	<u>54.55</u>	55.84
MixedShapesRegularTrain	68.33	74.56	94.64	88.82	<u>90.93</u>	88.45
MixedShapesSmallTrain	67.09	62.64	85.77	<u>82.43</u>	81.98	78.35
MoteStrain	81.31	80.51	86.18	86.42	90.65	<u>87.78</u>
NonInvasiveFetalECGThorax1	83.26	82.90	92.88	91.91	<u>92.72</u>	91.76
NonInvasiveFetalECGThorax2	86.31	86.92	93.28	<u>93.54</u>	93.94	92.98
OSULeaf	70.66	73.14	91.74	83.47	<u>86.78</u>	84.71
OliveOil	80.00	80.00	83.33	76.67	76.67	83.33
PLAID	50.09	41.71	43.39	43.95	<u>46.74</u>	45.25
PhalangesOutlinesCorrect	75.29	75.29	<u>79.49</u>	82.17	78.90	78.21
Phoneme	23.15	24.47	29.32	<u>29.91</u>	30.22	29.11
PickupGestureWiimoteZ	62.00	66.00	86.00	<u>82.00</u>	<u>82.00</u>	80.00
PigAirwayPressure	33.17	36.06	67.31	39.90	40.38	<u>42.79</u>
PigArtPressure	46.63	32.69	97.60	70.67	<u>86.06</u>	68.75
PigCVP	62.50	53.85	93.75	<u>82.69</u>	73.56	77.88
Plane	100.00	100.00	100.00	99.05	100.00	100.00
PowerCons	88.89	85.00	<u>95.56</u>	93.89	94.44	96.11
ProximalPhalanxOutlineAgeGroup	82.93	83.41	83.41	82.44	85.37	<u>84.39</u>
ProximalPhalanxOutlineCorrect	84.54	84.88	91.41	<u>89.00</u>	87.29	86.25
ProximalPhalanxTW	<u>80.00</u>	80.98	78.05	77.07	76.59	75.12
RefrigerationDevices	51.20	57.87	50.93	<u>52.53</u>	49.60	49.33
Rock	50.00	<u>56.00</u>	62.00	32.00	46.00	40.00
ScreenType	45.60	46.93	54.67	46.67	<u>51.47</u>	49.60
SemgHandGenderCh2	82.17	80.50	87.00	80.83	<u>83.17</u>	80.50
SemgHandMovementCh2	44.89	42.22	58.00	54.00	<u>55.56</u>	54.44
SemgHandSubjectCh2	59.11	55.33	<u>66.89</u>	63.56	67.11	65.11
ShakeGestureWiimoteZ	84.00	86.00	84.00	96.00	<u>92.00</u>	88.00
ShapeletSim	92.78	86.11	92.22	<u>98.89</u>	99.44	93.89
ShapesAll	66.33	67.50	91.17	84.17	<u>85.83</u>	81.83
SmallKitchenAppliances	79.73	79.20	80.27	78.93	<u>82.40</u>	84.27
SmoothSubspace	<u>96.00</u>	<u>96.00</u>	92.67	94.67	94.00	96.67
SonyAIBORobotSurface1	88.35	86.52	83.03	85.86	88.35	84.19
SonyAIBORobotSurface2	83.32	85.31	<u>92.97</u>	94.02	88.46	88.25
StarLightCurves	90.66	90.24	97.27	96.18	<u>96.39</u>	95.90
Strawberry	91.08	90.27	96.49	95.14	95.41	<u>95.68</u>
SwedishLeaf	84.64	85.76	<u>94.72</u>	95.04	94.24	93.92
Symbols	65.23	67.04	93.37	84.82	<u>92.96</u>	86.13
SyntheticControl	96.67	96.33	<u>97.33</u>	<u>97.33</u>	<u>97.33</u>	98.00
ToeSegmentation1	86.40	90.79	<u>93.42</u>	90.35	95.18	92.54
ToeSegmentation2	81.54	79.23	84.62	84.62	<u>86.92</u>	87.69
Trace	96.00	98.00	99.00	99.00	100.00	100.00
TwoLeadECG	<u>96.22</u>	96.58	89.73	94.47	83.93	94.29
TwoPatterns	95.33	96.08	98.62	98.97	97.88	<u>98.83</u>

UMD	82.64	72.92	97.92	97.22	96.53	97.92
UWaveGestureLibraryAll	60.69	61.17	82.58	77.53	<u>78.95</u>	77.22
UWaveGestureLibraryX	64.63	65.02	76.35	75.77	<u>76.21</u>	74.48
UWaveGestureLibraryY	51.81	53.13	66.39	<u>65.44</u>	<u>65.44</u>	64.21
UWaveGestureLibraryZ	62.65	63.07	<u>72.84</u>	72.86	71.08	71.55
Wafer	96.37	96.66	<u>99.19</u>	99.16	99.27	99.11
Wine	79.63	<u>83.33</u>	<u>83.33</u>	79.63	<u>83.33</u>	85.19
WordSynonyms	44.36	41.38	62.23	<u>61.76</u>	60.34	<u>61.76</u>
Worms	59.74	58.44	70.13	70.13	70.13	66.23
WormsTwoClass	67.53	64.94	<u>75.32</u>	80.52	70.13	70.13
Yoga	67.37	67.63	83.10	79.10	<u>82.20</u>	79.63
Average	71.93	71.64	80.99	78.76	79.23	79.05

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